

Homework 4

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1.

3.5)

We are tasked with showing that $x_t - x_{t-1} = \delta + w_t + y_t - y_{t-1}$ is stationary and w_t and independent of y_t .

This is assuming that the model is $x_t = \mu_t + y_t$, with y_t a stationary process.

For a time series to be stationary it must have a constant mean that does not depend on t and a covariance function that only depends on lag h .

We first look at the constant mean that does not depend on t :

$$E[x_t - x_{t-1}] = E[\delta + w_t + y_t - y_{t-1}] = \delta + E[y_t] - E[y_{t-1}]$$

Then, as y_t, y_{t-1} are stationary processes their expected value would be constant and not depend on t .

Next we look at the covariance function.

$$\gamma(0) = \text{Var}(x_t - x_{t-1}) = \text{Var}(\delta + w_t + y_t - y_{t-1}) = \text{Var}(w_t) + \text{Var}(y_t) + \text{Var}(y_{t-1}) - 2\text{Cov}(y_t, y_{t-1})$$

Since y_t, y_{t-1} are stationary processes their covariance depends on lag h . So:

$$\gamma(0) = \sigma_w^2 + 2\gamma_y(0) - 2\gamma_y(1)$$

$$\gamma(1) = \text{Cov}(x_t - x_{t-1}, x_{t+1} - x_t) = \text{Cov}(\delta + w_t + y_t - y_{t-1}, \delta + w_{t+1} + y_{t+1} - y_t)$$

$$= Cov(y_t, y_{t+1}) - Cov(y_t, y_t) - Cov(y_{t-1}, y_{t+1}) + Cov(y_{t-1}, y_t) = \gamma_y(1) - \gamma_y(0) - \gamma_y(2) + \gamma_y(1)$$

$$= 2\gamma_y(1) - \gamma_y(0) - \gamma_y(2)$$

We can see that $\gamma(1)$ also depends only on lag h .

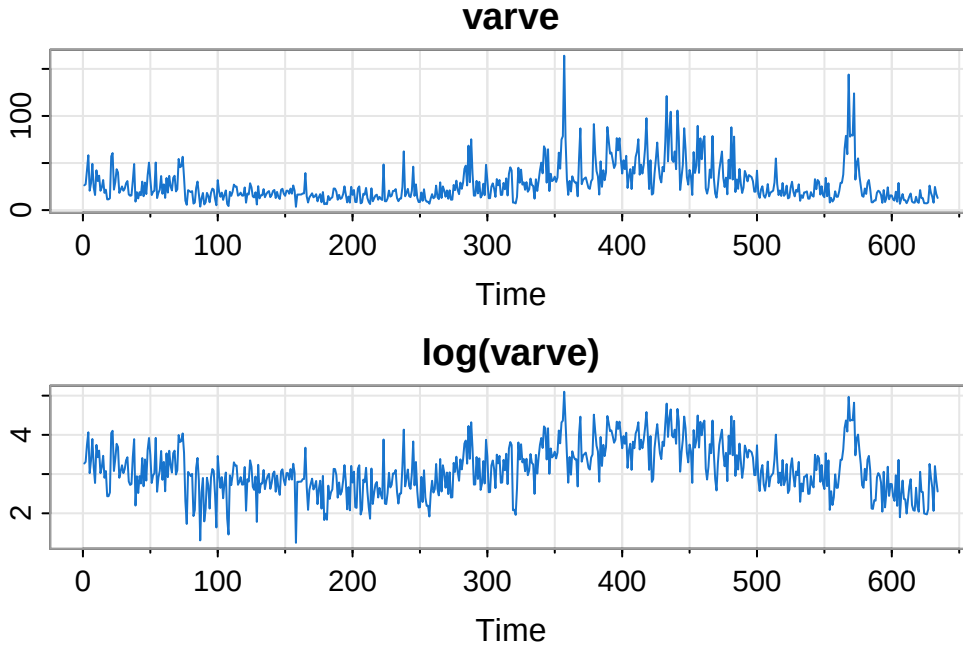
This pattern continues on, with $\gamma(h)$ depending only on linear combinations of $\gamma_y(h)$ and thus depending only on h .

With this we can say $x_t - x_{t-1}$ is a stationary process.

3.6)

The sample variance in the first half of the data is 133.457 , in the second half of the data it is 592.964. The large difference shows evidence of non constant variable, it seems variance is getting larger as time passes.

A common technique to address non-constant variance is to apply a log transformation. This is attempted below:

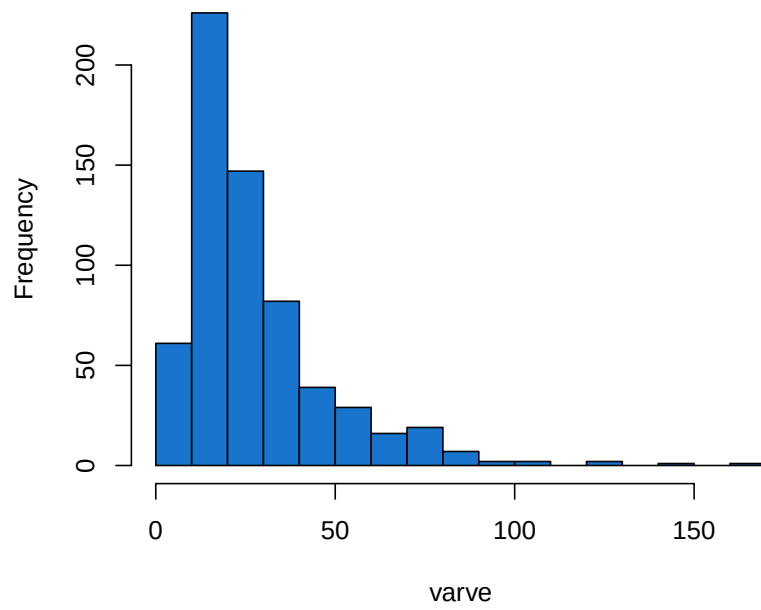


In the original Varve data there is strong evidence of non-constant variance visually, it can easily be seen that early oscillations have much smaller magnitude than later ones.

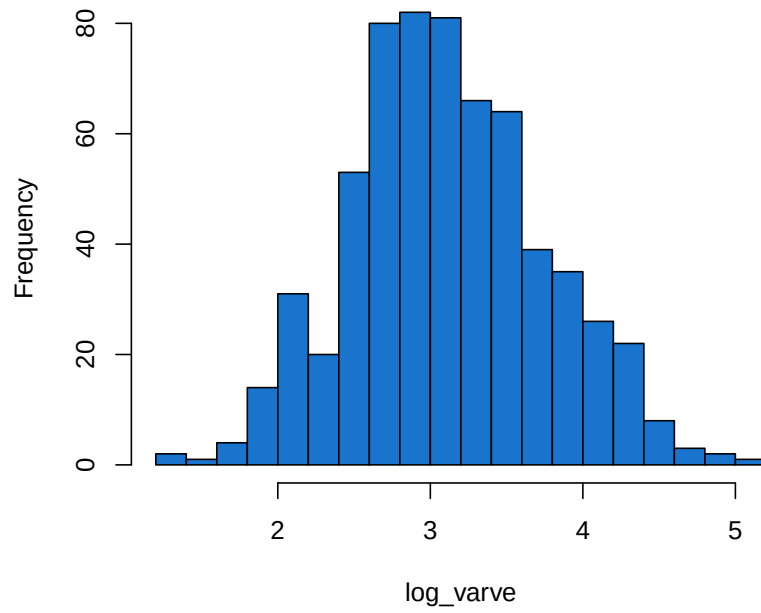
The log transformation brings the oscillation magnitudes closer, approaching constant variance.

A log transformation can also assist the transformation to normality.

Histogram of varve

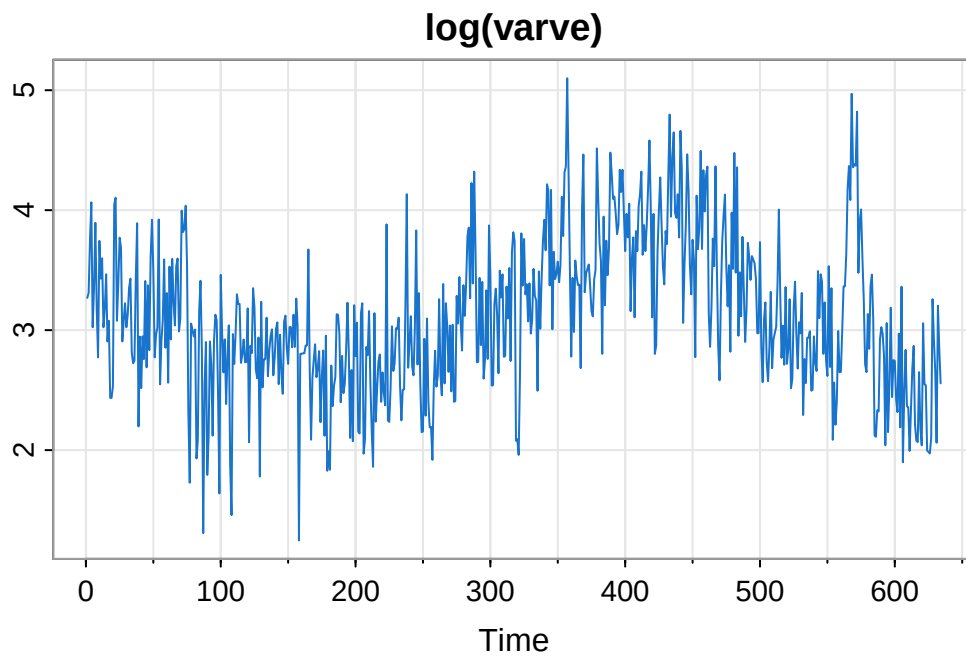


Histogram of log(varve)

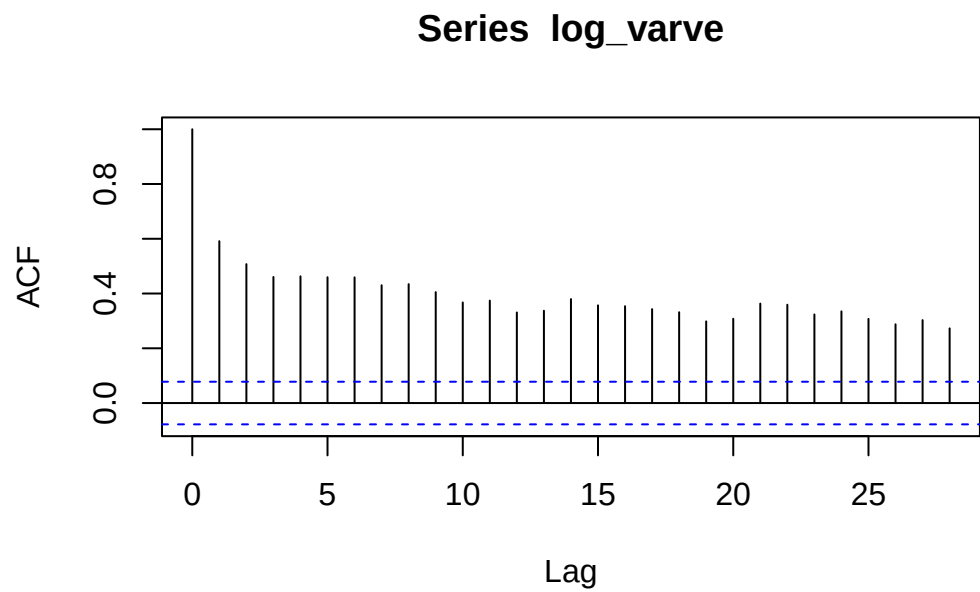


The Varve data exhibits a long right tail. The log transformed varve data is close to symmetrical, moving it closer to normality.

b)

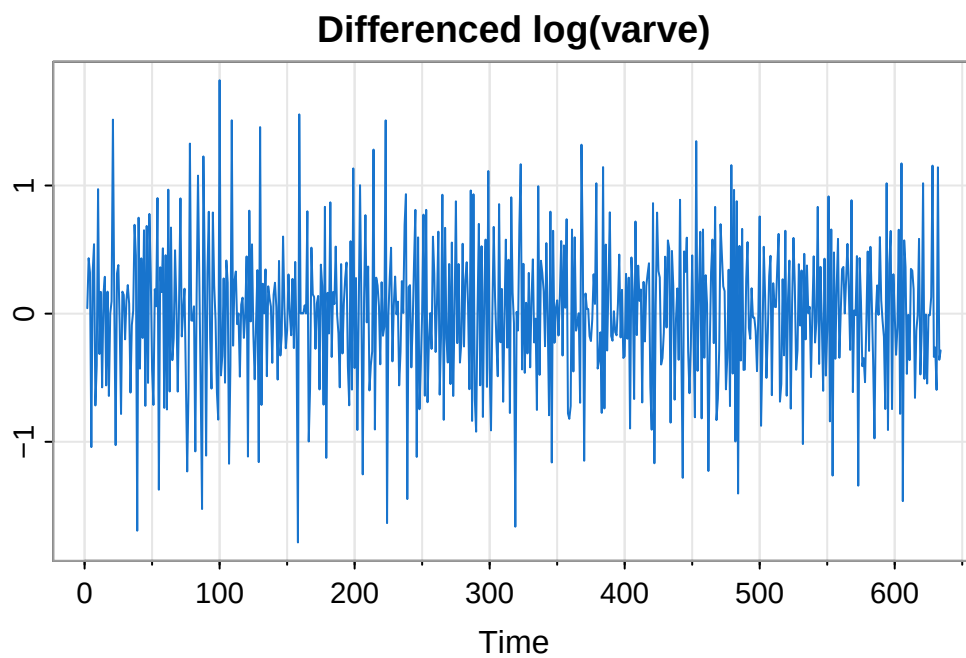


From time period 200 to 400 we can see a similar upward trend to the temperatures in figure 1.9.

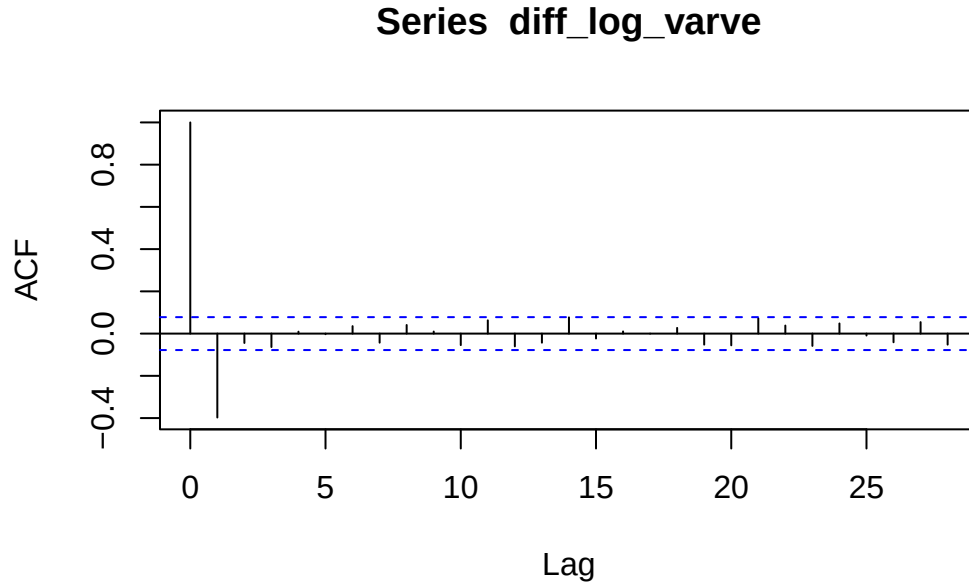


The sample ACF shows evidence of prolonged positive correlation that trends to zero very slowly.

d)



When examining the time series plot of the differenced log transformed varve we see fast choppy oscillations with no evidence of a trend or correlation. This plot looks similar to previous stationary series' that we have analyzed in the past.

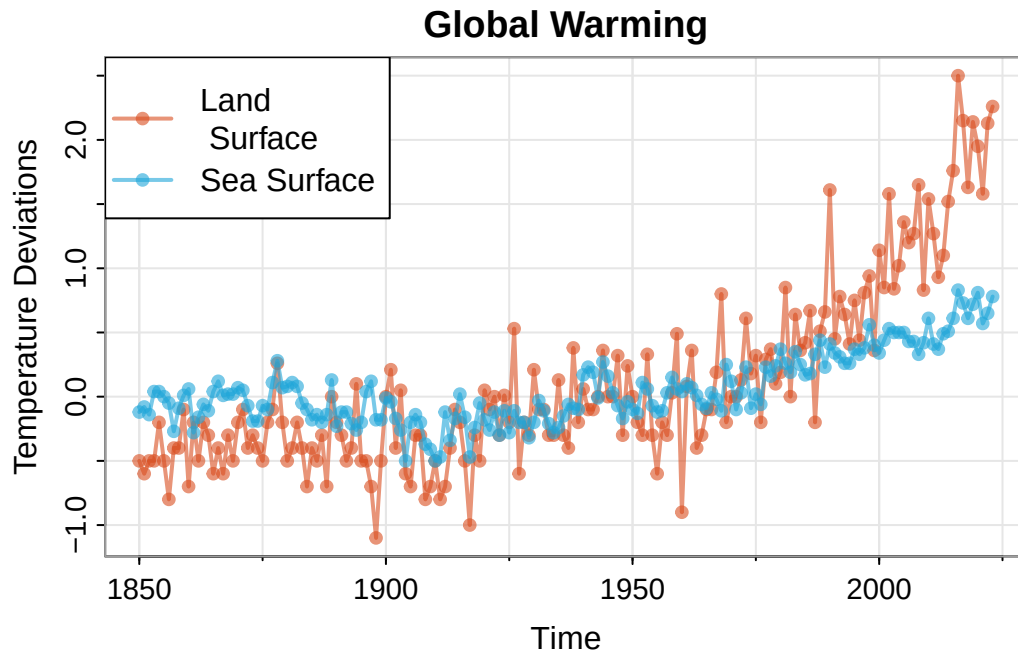


The sample ACF plot is consistent with the previous plot, the only lag with a significant auto-correlation is lag 1, after which the correlation quickly trends to zero consistent with the behavior we would expect in a stationary series.

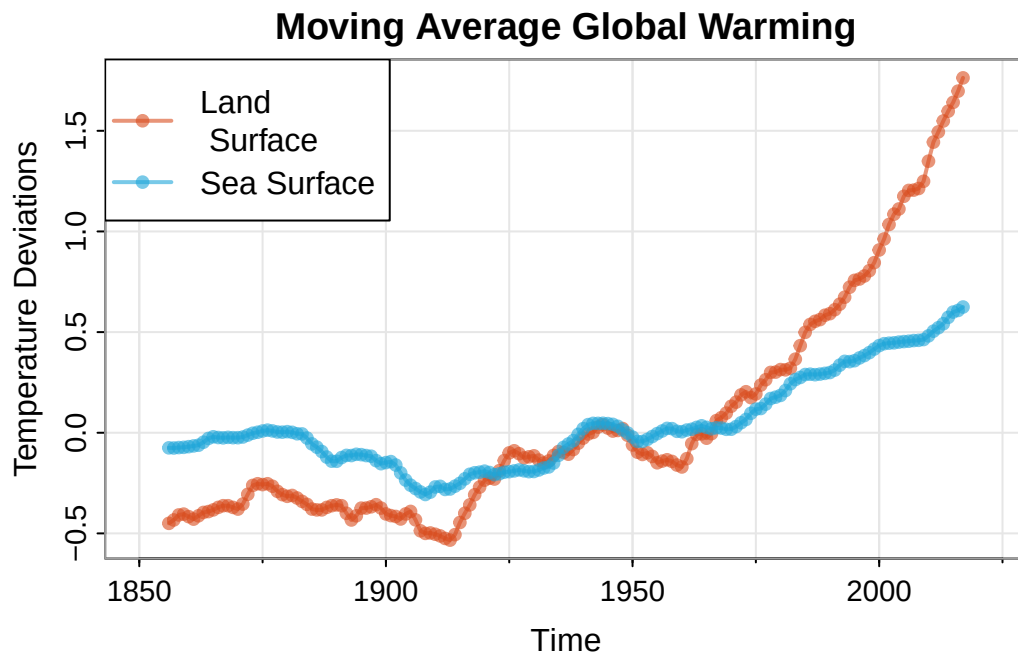
The differenced log transformed varve data can be interpreted as the year over year percentage change in global varve thickness from Massachusetts . This is because of the below:

$$\log(y_t) - \log(y_{t-1}) = \log(y_t/y_{t-1}) \approx (y_t - y_{t-1})/y_{t-1}$$

3.7)



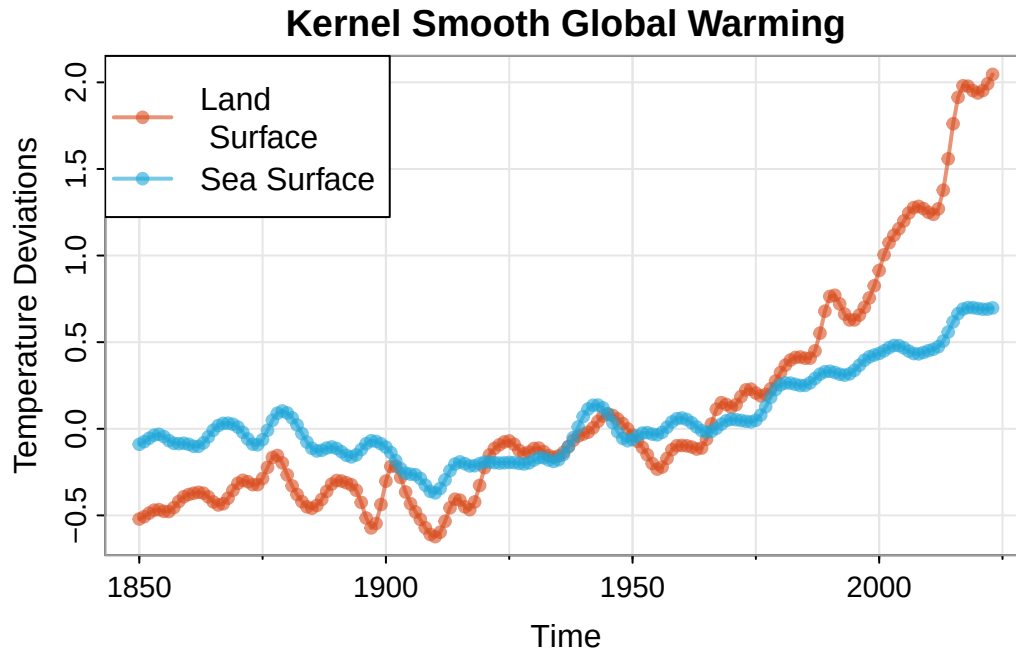
We start with example 3.16 which used the moving average smoother. This used weights of $1/12$ for all months but $1/24$ th at the end so the same month doesn't get double counted.



Compared to the original time series plot this is much smoother, removing the choppiness and

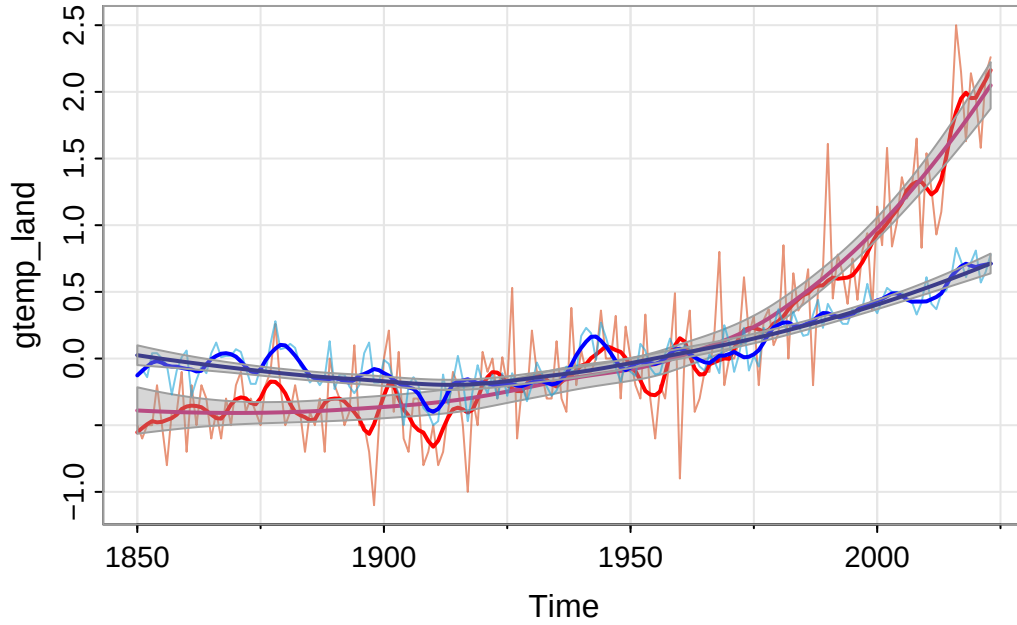
making it easier to see the trends in the data.

Example 3.17 used kernel smoothing. Kernel smoothing uses a weight function to average observations, the selection of bandwidth was 5 was chosen which would look at smoothing over a half decade.



This plot introduces more choppiness compared to the moving average smoothing, we can see the short term deviations more clearly. It still makes it easier to see the overall trends when compared to the original plot.

Example 3.18 uses the Lowess method, which is similar to nearest neighborhood regression. In line with the example we use the 5% of data and then the default 2/3rd of data.



We can see that when the loess has a small smoother span like 5% it performs very similarly to the previous two approaches. When it is set to a large amount of the data, like the default of 2/3, it shows a very smooth trend that ignores most local oscillations.

4.2)

We have the model $x_t = \phi x_{t-1} + w_t$ with $w_t \sim wn(0, \sigma_w^2)$ and $|\phi| < 1$ and constant and $x_0 = w_0$.

a)

We substitute in x_{t-1} :

$$x_t = \phi(\phi x_{t-2} + w_{t-1}) + w_t = \phi^2 x_{t-2} + \phi w_{t-1} + w_t$$

We substitute in x_{t-2} :

$$x_t = \phi^2(\phi x_{t-3} + w_{t-2}) + \phi w_{t-1} + w_t = \phi^3 x_{t-3} + \phi^2 w_{t-2} + \phi w_{t-1} + w_t$$

Then, since $x_0 = w_0$, we can say:

$$x_t = \sum_{j=0}^t \phi^j w_{t-j}$$

b)

$$E[x_t] = E\left[\sum_{j=0}^t \phi^j w_{t-j}\right] = \sum_{j=0}^t \phi^j E[w_{t-j}] = 0$$

c)

$$Var(x_t) = Var\left(\sum_{j=0}^t \phi^j w_{t-j}\right) = \sum_{j=0}^t \phi^{2j} Var(w_{t-j}) = \sum_{j=0}^t \phi^{2j} \sigma_w^2$$

We recognize this as a geometric series. We know the below:

$$a \sum_{j=0}^t r^j = a(1 - r^{t+1})/(1 - r)$$

So:

$$\sigma_w^2 \sum_{j=0}^t \phi^{2j} = \sigma_w^2 \frac{1 - \phi^{2(t+1)}}{1 - \phi^2}$$

d)

Starting with $h = 1$ since we already found $Var(x_t)$:

$$\gamma(1) = Cov(x_{t+1}, x_t) = Cov(\phi x_t + w_{t+1}, x_t) = \phi Var(x_t)$$

With $h = 2$:

$$\gamma(2) = Cov(x_{t+2}, x_t) = Cov(\phi x_{t+1} + w_{t+2}, x_t) = Cov(\phi(\phi x_t + w_{t+1}) + w_{t+2}, x_t) = Cov(\phi^2 x_t + \phi w_{t+1} + w_{t+2}, x_t) = \phi^2 Var(x_t)$$

The pattern is clear, this leaves us with:

$$\gamma(h) = \phi^h Var(x_t)$$

e)

Since $Var(x_t)$ depends on t , x_t is not stationary.

f)

As $t \rightarrow \infty$ we have:

$$\lim_{t \rightarrow \infty} Var(x_t) = \frac{\sigma_w^2}{1 - \phi^2} (1 - \lim_{t \rightarrow \infty} \phi^{2(t+1)})$$

Since $|\phi| < 1$ as t tends towards infinity $\lim_{t \rightarrow \infty} \phi^{2(t+1)} = 0$. So asymptotically we would have:

$$Var(x_t) = \frac{\sigma_w^2}{1 - \phi^2}$$

And also:

$$\phi(h) = \phi^h \frac{\sigma_w^2}{1 - \phi^2}$$

Which depends only on lag h . So as t approaches infinity the auto-covariance function of x_t tends toward only depending on h . In that way x_t is asymptotically stationary.

g)

We could use the simulated i.i.d values by using the form $x_t = \sum_{j=0}^t \phi^j w_{t-j}$ and selecting a certain ϕ value. Then the cumulative sum would be the n observations of the stationary AR(1) model, provided n is large to invoke the asymptotic stationary result.

h)

If $x_0 = w_0 / \sqrt{1 - \phi^2}$ then:

$$Var(x_0) = \sigma_w^2 / (1 - \phi^2)$$

Then $x_1 = \phi x_0 + w_1$ which would have variance:

$$Var(x_1) = \phi^2 \frac{\sigma_w^2}{1 - \phi^2} + \sigma_w^2 = \frac{\phi^2 \sigma_w^2 + (1 - \phi^2) \sigma_w^2}{1 - \phi^2} = \frac{\sigma_w^2}{1 - \phi^2}$$

This pattern continues, so $Var(x_t)$ does not depend on t . Since the auto-covariance function is:

$$\gamma(h) = \phi^h Var(x_t)$$

We can say that $\gamma(h)$ does not depend on t , as $Var(x_t)$ does not depend on t . With this restriction x_t is now stationary.

4.3)

We have two models:

$$(i) \quad x_t = .80x_{t-1} - .15x_{t-2} + w_t - .30w_{t-1}$$

$$(ii) \quad x_t = x_{t-1} - .50x_{t-2} + w_t - w_{t-1}$$

a)

We can show the concise form of both models:

$$(i) \quad (1 - .8B + .15B^2)x_t = (1 - .3B)w_t$$

$$(ii) \quad (1 - 1B + .5B^2)x_t = (1 - 1B)w_t$$

We check the redundancy using R:

i)

```
[1] 2.000000 3.333333
```

```
[1] 3.333333
```

The root of 3.33 indicates there is parameter redundancy.

We can attempt to simplify the and find the like roots:

$$(1 - .8B + .15B^2)x_t = (1 - .3B)w_t \quad (1 - .3B)(1 - .5B)x_t = (1 - .3B)w_t$$

So we can represent the model as:

$$(1 - .5B)x_t = (1 - .3B)w_t$$

Or:

$$x_t = 0.5x_{t-1} + w_t - 0.3w_{t-1}$$

ii)

```
[1] 1.414214 1.414214
```

```
[1] 1
```

The roots are not close, this indicates there is no redundancy.

b)

Model i was reduced, so we can use R to compute the roots again:

```
[1] 2
```

```
[1] 3.333333
```

The root of the AR term are greater than 1 so this is a casual model.

The root of the MA term are greater than 1 so this is an invertible model.

```
[1] 1.414214 1.414214
```

```
[1] 1
```

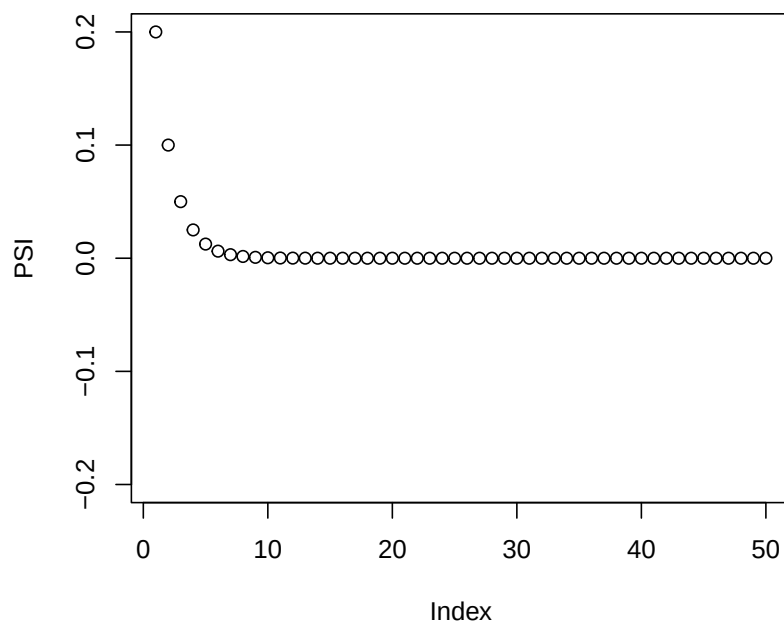
The roots of the AR term is greater than 1, so this is a casual model.

The root of the MA term is equal to 1, so this is not an invertible model.

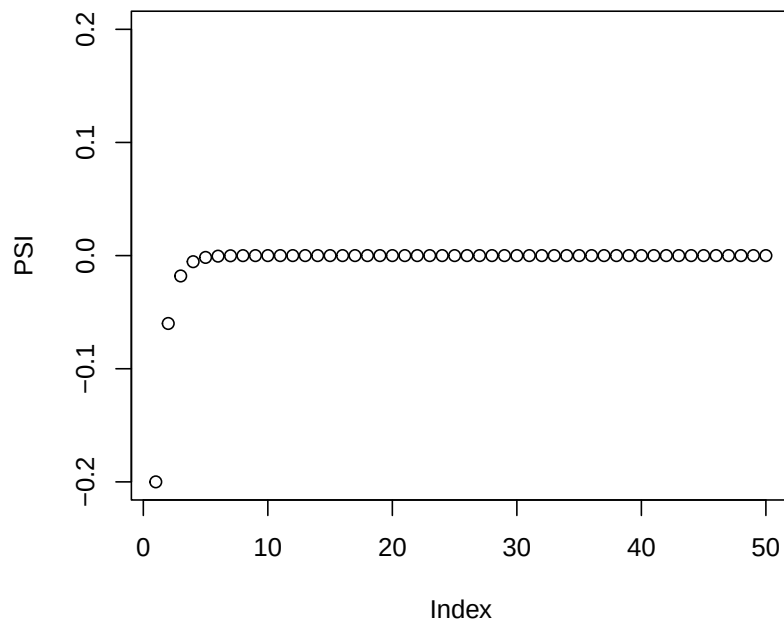
c)

Starting with model 1:

Causal Check



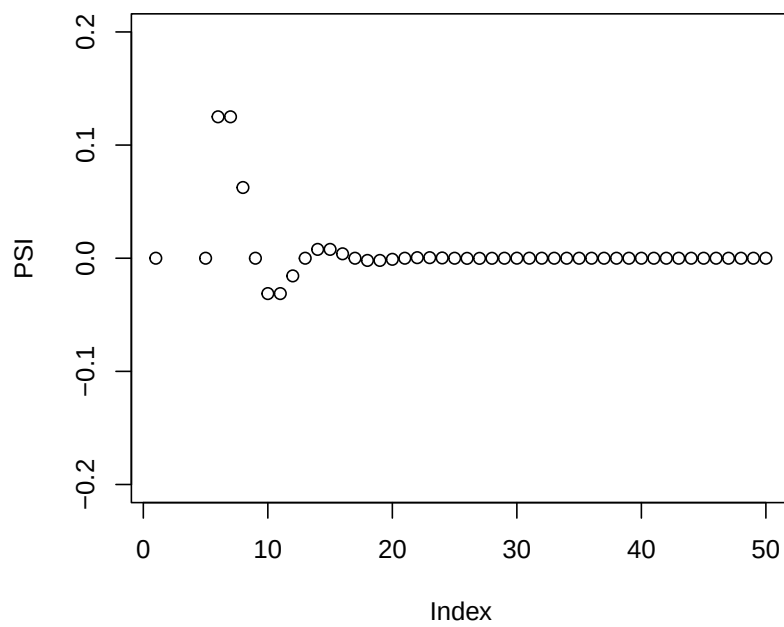
Invertible Check



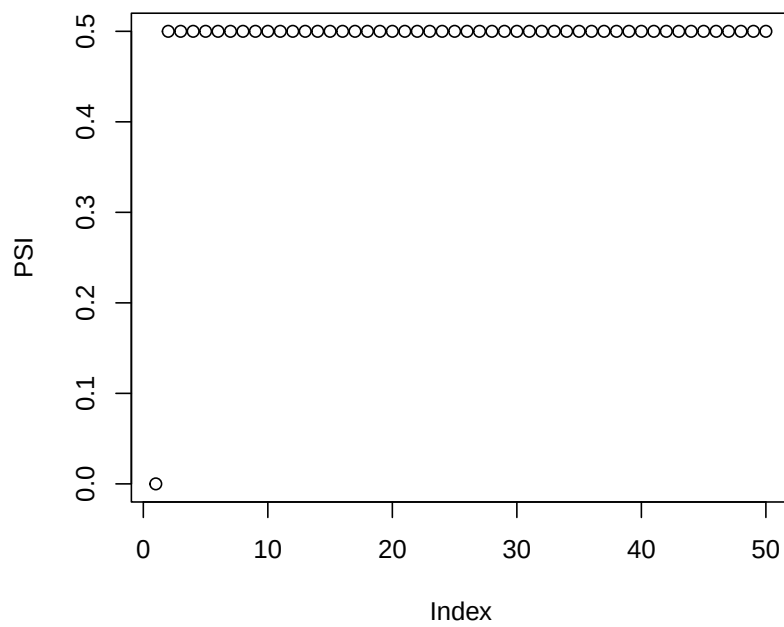
We can see that both tend to zero, so this model is indeed casual and invertible.

Then model 2:

Causal Check

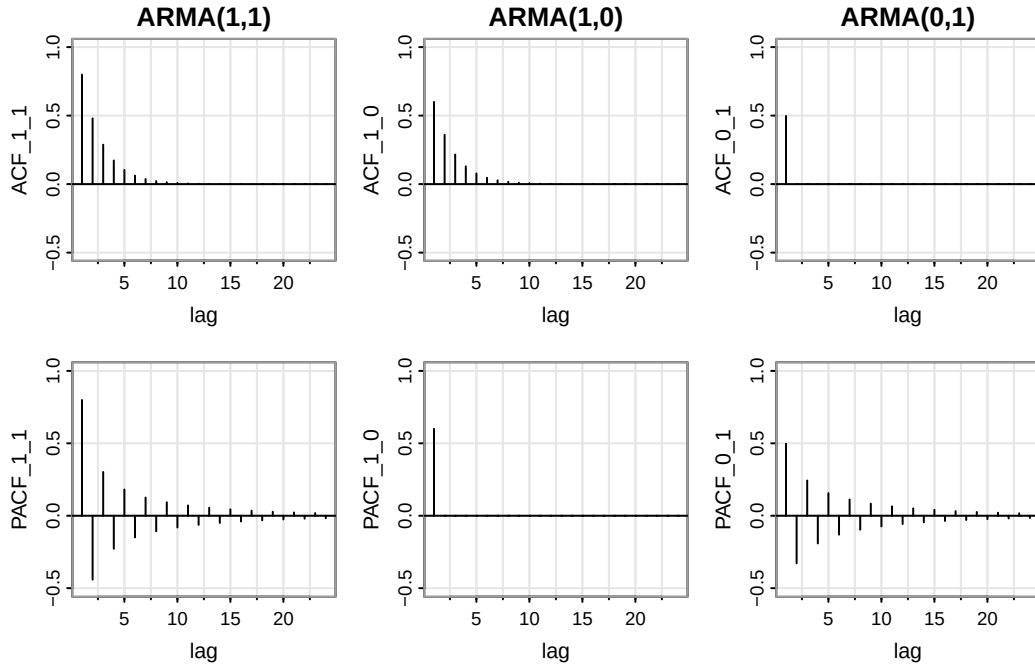


Invertible Check



The weights tend to zero in the casual check. They don't tend to zero in the invertible check, verifying our previous conclusion that model ii is not invertible.

4.4)

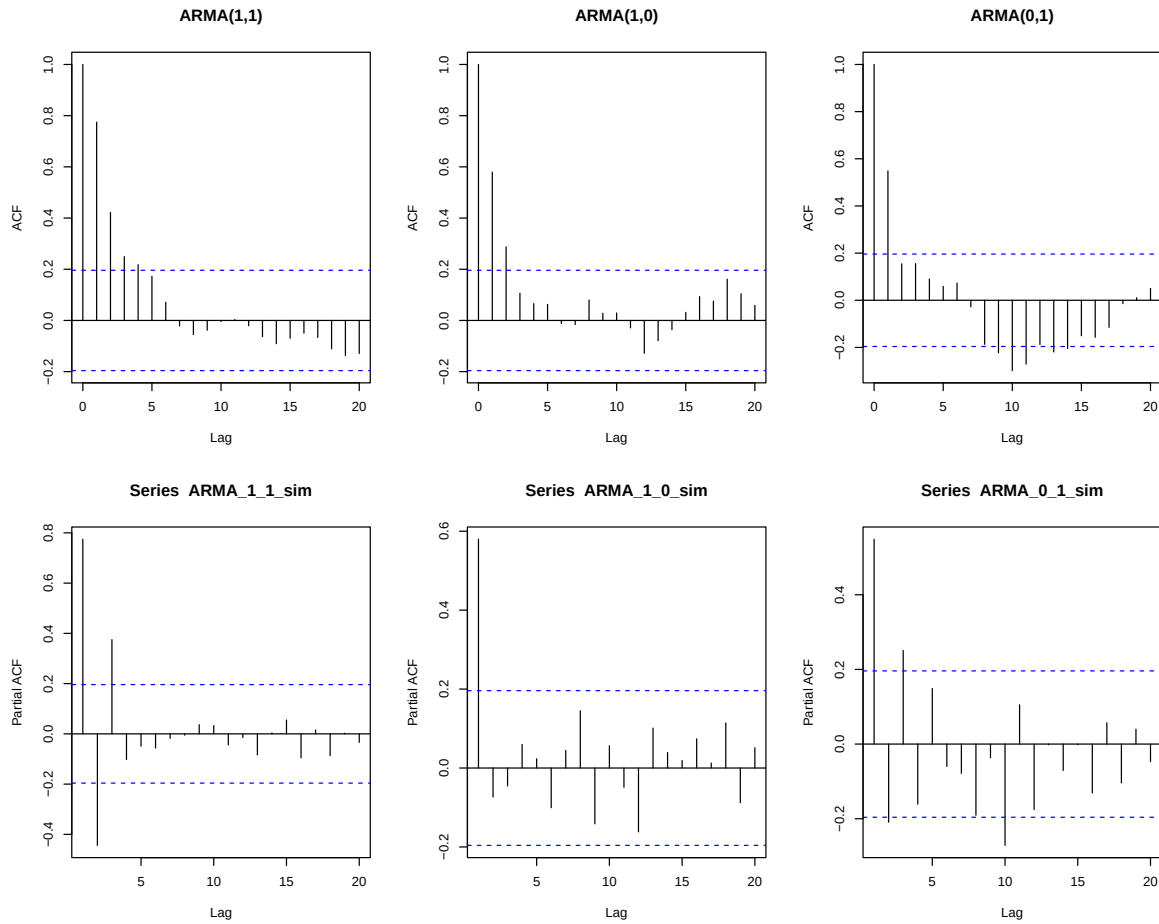


The ARMA(1,1) model has an ACF plot that tends toward 0 over time and a PACF that alternates positive and negative that tends toward zero.

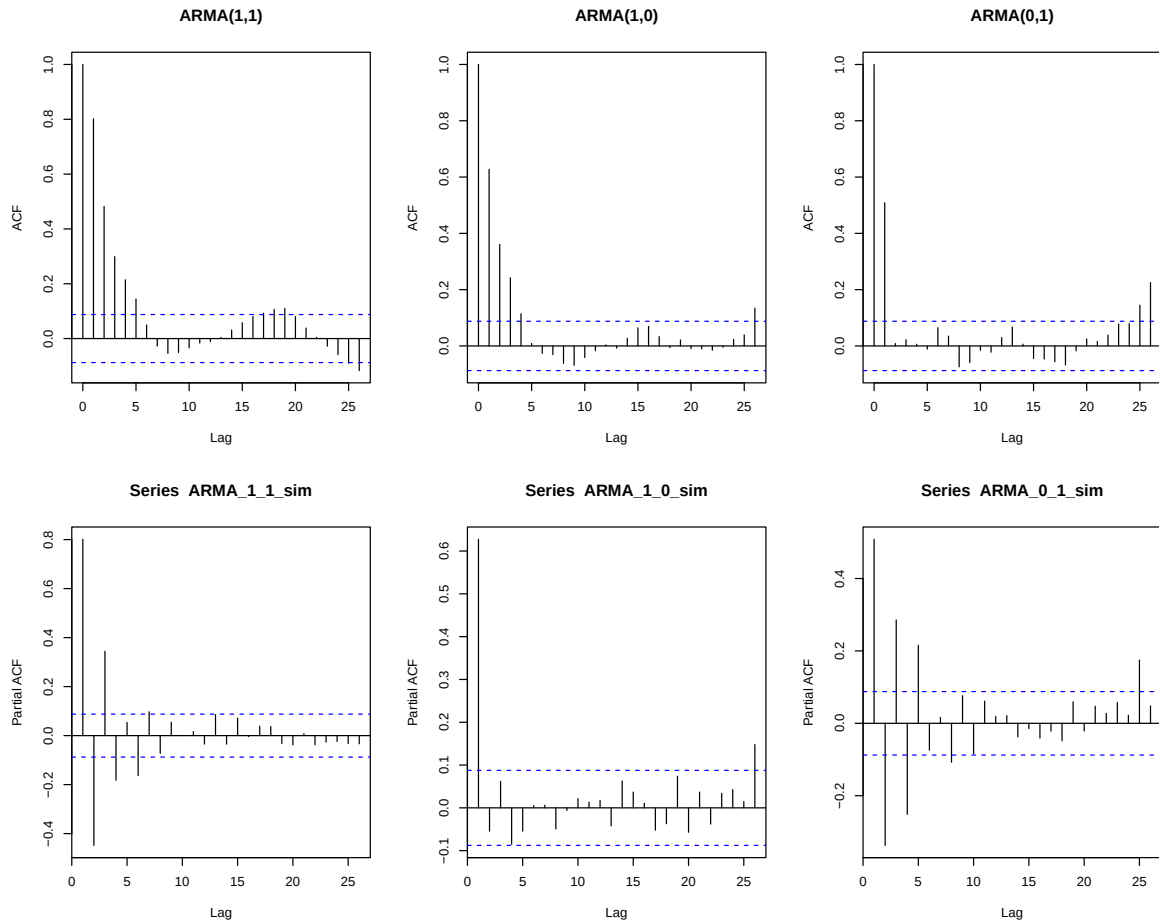
The ARMA(1,0) model has a similar ACF plot as ARMA(1,1), tending to 0 over time but with a smaller magnitude at all lags. The PACF that has correlation at lag 1 but none at later lags.

The ARMA(0,1) model has a ACF plot that has a positive value at lag 1 but none at further lags. the PACF has alternating positive and negative correlations that tend toward zero but at a lower magnitude when compared to ARMA(1,1).

Recognizing these patterns would allow us to choose the correct model from sample data.



The sample ACF and PACFs are pretty similar to the theoretical values, there are a few cases where there are lags that are significant that we wouldn't expect. In particular ARMA(1,0) has lags that are significant in the negative direction in lag 15 that we wouldn't expect as it should tend to zero. The same idea can be seen in AR(0,1) we expect all those lags to be zero past lag 1 but we see significance in lag 17.



With $n = 500$ the patterns are closer to the theoretical values but we see more correlations poking into significance, likely due to the increased power from increasing n .

II.

a)

Stationarity is important because the covariance and mean functions can be estimated by averaging. Confidence intervals can be constructed using the CLT.

b)

MA models are important because they allow us to estimate the weights of a filter of weight noise terms, helpful for any stationary time series.

c)

AR models are important because we can estimate weights of a time series and ensure the present shock doesn't depend on future data.

III.

a)

$$E[x_t] = E[w_{t+1} + \sum_{j=0}^{\infty} \phi^j w_{t-j}] = 0$$

$$Cov(x_t, x_t) = Var(x_t) = \sigma_w^2 + \sum_{j=0}^{\infty} \phi^{2j} \sigma_w^2 = \sigma_w^2 + \frac{\sigma_w^2}{1 - \phi^2}$$

$$Cov(x_{t+1}, x_t) = Cov(w_{t+2} + \sum_{j=0}^{\infty} \phi^j w_{t+1-j}, w_{t+1} + \sum_{k=0}^{\infty} \phi^k w_{t-k})$$

$$= Cov(w_{t+2}, \sum_{k=0}^{\infty} \phi^k w_{t-k}) + Cov(\sum_{j=0}^n \phi^j w_{t+1-j}, \sum_{k=0}^{\infty} \phi^k w_{t-k}) + Cov(\sum_{j=0}^{\infty} \phi^j w_{t+1-j}, w_{t+1})$$

$$= 0 + \frac{\sigma_w^2 \phi}{1 - \phi^2} + \sigma_w^2$$

Following that pattern, if we had $t + 2$ we would have:

$$Cov(x_{t+2}, x_t) = Cov(\sum_{j=0}^n \phi^j w_{t+2-j}, \sum_{k=0}^{\infty} \phi^k w_{t-k}) + Cov(\sum_{j=0}^{\infty} \phi^j w_{t+2-j}, w_{t+1})$$

$$= \frac{\sigma_w^2 \phi^2}{1 - \phi^2} + \phi \sigma_w^2$$

So we can say that:

$$\gamma(h) = \frac{\phi^h \sigma_w^2}{1 - \phi^2} + \phi^h \sigma_w^2$$

b)

For a process to be stationary its mean must be constant and not depend on t as well as having a covariance function that does not depend on t .

Since the mean function is constant and does not depend on t and the autocovariance function does not depend on t , and only on lag h we can say this is stationary.

c)

This process is clearly not casual because x_t is directly dependent on a future shock w_{t+1} .

Multiple Choice

1)

e

2)

c

3)

c